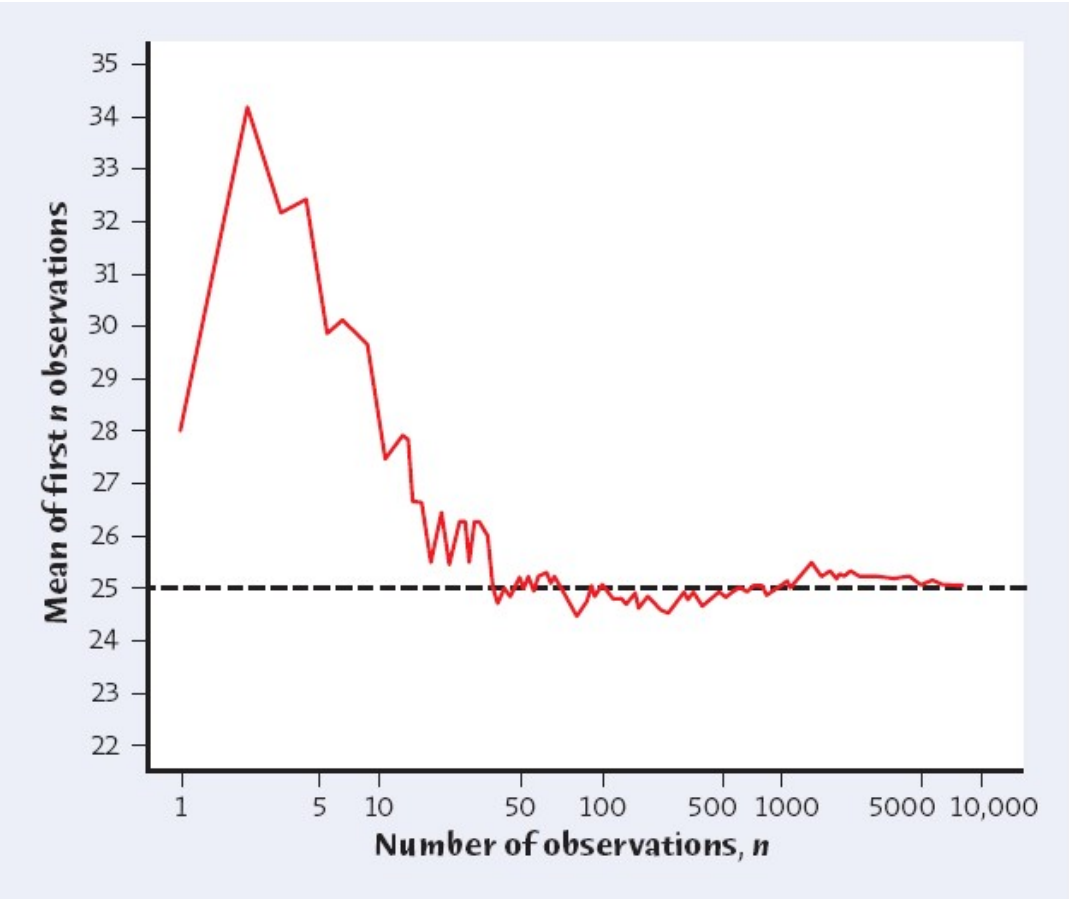
Sampling Distribution

LAW OF LARGE NUMBERS

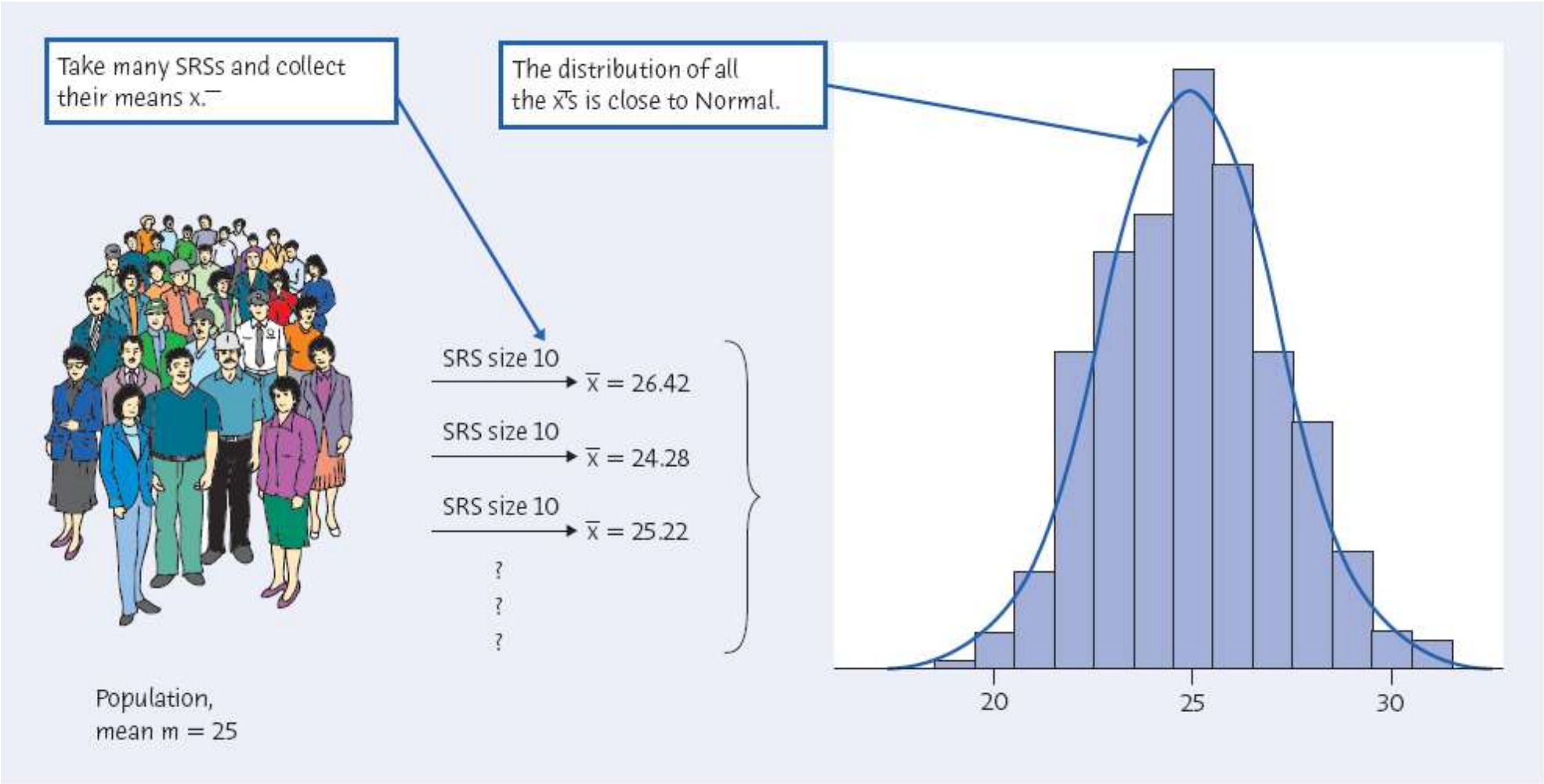
Draw observations at random from any population with finite mean μ. As the number of observations drawn increases, the mean $\bar{x}$ of the observed values gets closer and closer to the mean μ of the population.



Sampling Distribution

- Take a large number of samples of size 10 from the population.
- Calculate the sample mean $\overline{x}$ for each sample.
- Make a histogram of the values of $\overline{x}$.
- Examine the shape, center, and spread of the distribution displayed in the histogram.

Sampling Distribution



Sampling Distribution

POPULATION DISTRIBUTION, SAMPLING DISTRIBUTION

The **population distribution** of a variable is the distribution of values of the variable among all the individuals in the population.

The **sampling distribution** of a statistic is the distribution of values taken by the statistic in all possible samples of the same size from the same population.

MEAN AND STANDARD DEVIATION OF A SAMPLE MEAN[2]

Suppose that $\bar{x}$ is the mean of an SRS of size n drawn from a large population with mean μ and standard deviation σ. Then the sampling distribution of $\bar{x}$ has mean μ and **standard deviation** $\sigma/\sqrt{n}$.

$$\mathrm{Var}(\bar{x}) = \mathrm{Var}\left(\frac{1}{n}\sum X_i\right) = \frac{1}{n^2}\sum \mathrm{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

$$\mathrm{SD}(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

Sampling Distribution

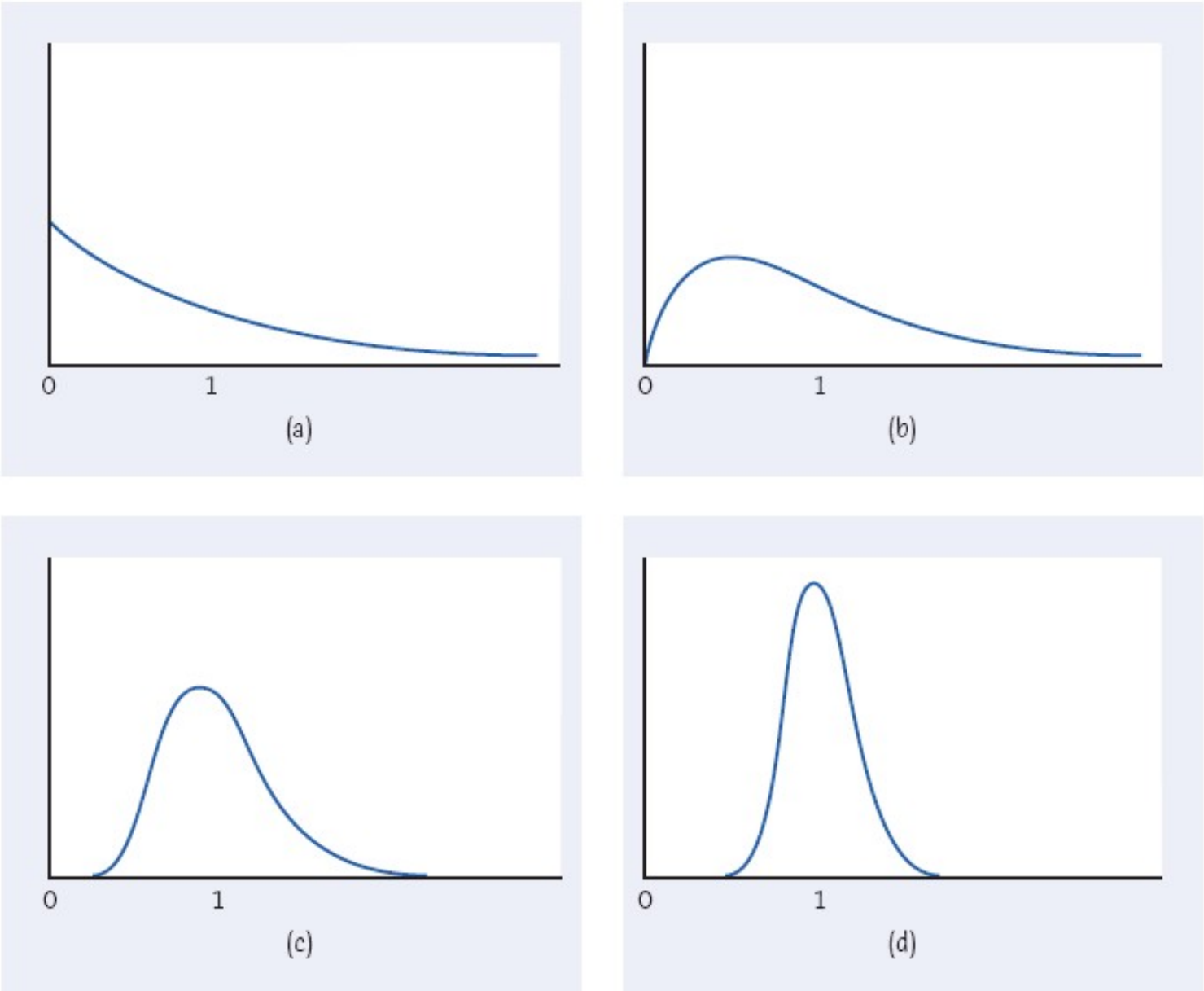
CENTRAL LIMIT THEOREM

Draw an SRS of size n from any population with mean μ and finite standard deviation σ. The **central limit theorem** says that when n is large, the sampling distribution of the sample mean $\bar{x}$ is approximately Normal:

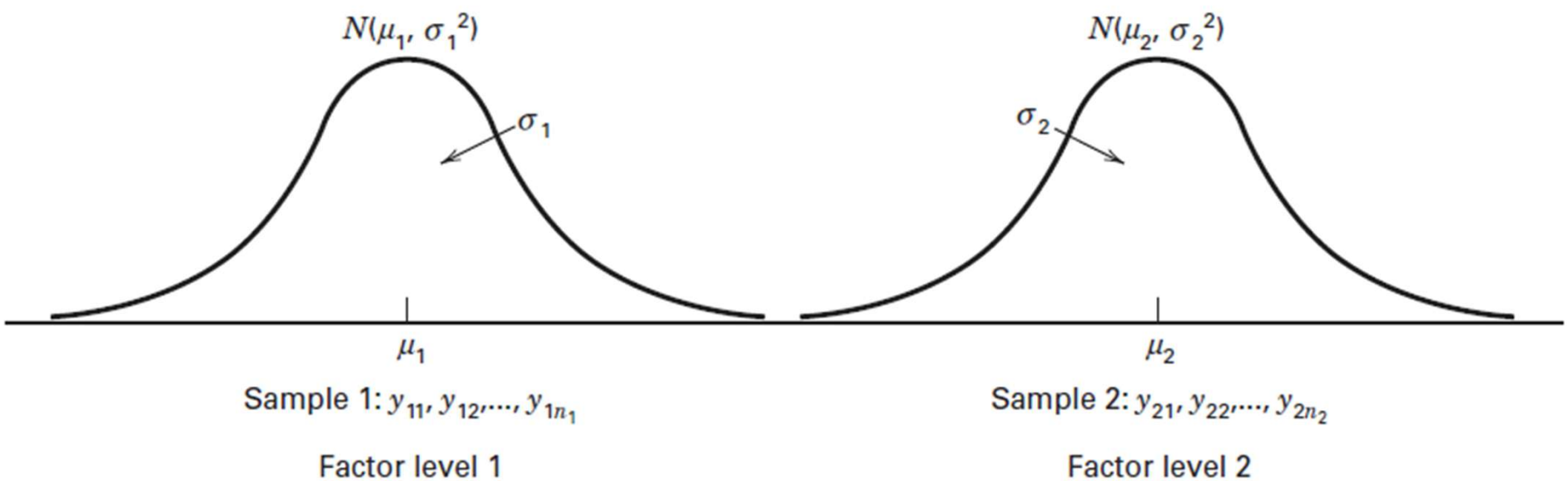
$$\bar{x} \text{ is approximately } N\left(\mu, \frac{\sigma^2}{n}\right)$$

The central limit theorem allows us to use Normal probability calculations to answer questions about sample means from many observations even when the population distribution is not Normal.

Central Limit Theorem



Hypothesis Testing



A Model for the Data. We often describe the results of an experiment with a **model**. A simple statistical model that describes the data from an experiment such as we have just described is

$$y_{ij} = \mu_i + \epsilon_{ij} \begin{cases} i = 1, 2 \\ j = 1, 2, \ldots, n_i \end{cases} \tag{2.23}$$

where y_{ij} is the jth observation from factor level i, μ_i is the mean of the response at the ith factor level, and ϵ_{ij} is a normal random variable associated with the ijth observation.

Hypothesis Testing

Statistical Hypotheses. A **statistical hypothesis** is a statement either about the parameters of a probability distribution or the parameters of a model. The hypothesis reflects some **conjecture** about the problem situation. For example, in the Portland cement experiment, we may think that the mean tension bond strengths of the two mortar formulations are equal. This may be stated formally as

$$H_0: \mu_1 = \mu_2$$
$$H_1: \mu_1 \neq \mu_2$$

where μ_1 is the mean tension bond strength of the modified mortar and μ_2 is the mean tension bond strength of the unmodified mortar. The statement $H_0: \mu_1 = \mu_2$ is called the **null hypothesis** and $H_1: \mu_1 \neq \mu_2$ is called the **alternative hypothesis**. The alternative hypothesis specified here is called a **two-sided alternative hypothesis** because it would be true if $\mu_1 < \mu_2$ or if $\mu_1 > \mu_2$.

Hypothesis Testing

In hypothesis testing you always face two possibilities:

- The null hypothesis H_0 is **true**
- The null hypothesis H_0 is **false**

And you make one of two decisions:

- Reject H_0
- Do not reject H_0

This gives a **2×2 table**:

Reality	Your decision	Result
H_0 true	Reject	**Type I error**
H_0 true	Do not reject	Correct
H_0 false	Reject	Correct
H_0 false	Do not reject	**Type II error**

Hypothesis Testing

Two kinds of errors may be committed when testing hypotheses. If the null hypothesis is rejected when it is true, a type I error has occurred. If the null hypothesis is *not* rejected when it is false, a type II error has been made. The probabilities of these two errors are given special symbols

$$\alpha = P(\text{type I error}) = P(\text{reject } H_0 | H_0 \text{ is true})$$
$$\beta = P(\text{type II error}) = P(\text{fail to reject } H_0 | H_0 \text{ is false})$$

Sometimes it is more convenient to work with the **power** of the test, where

$$\text{Power} = 1 - \beta = P(\text{reject } H_0 | H_0 \text{ is false})$$

The general procedure in hypothesis testing is to specify a value of the probability of type I error α, often called the **significance level** of the test, and then design the test procedure so that the probability of type II error β has a suitably small value.

Power is the ability of a test to detect a real effect!

The Two-Sample t-Test

Objective:
To test whether **two population means are equal**:

$$H_0: \mu_1 = \mu_2 \quad \text{vs} \quad H_1: \mu_1 \neq \mu_2$$

Assumptions:

- Two independent random samples
- Both populations are approximately normal
- **Equal variances**: $\sigma_1^2 = \sigma_2^2 = \sigma^2$

Test Statistic:

$$t_0 = \frac{\bar{y}_1 - \bar{y}_2}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where
$\bar{y}_1$, $\bar{y}_2$ = sample means, n_1, n_2 = sample sizes, S_p = pooled standard deviation

The Two-Sample t-Test

Pooled Variance Estimate:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

This is the **best estimate of common variance**.

Standard Error of Difference:

$$SE(\bar{y}_1 - \bar{y}_2) = S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Measures **uncertainty in the mean difference**.

Decision Rule:

Compare $|t_0|$ with $t_{\alpha/2, n_1+n_2-2}$

If $|\boldsymbol{t_0}| > \boldsymbol{t_{\alpha/2, d_f}}$ $\Rightarrow$ **Reject** $\boldsymbol{H_0}$

The Two-Sample t-Test

Case 1: Testing $\mu_1 > \mu_2$

- Null hypothesis

$$H_0 : \mu_1 \leq \mu_2$$

- Alternative hypothesis

$$H_1 : \mu_1 > \mu_2$$

This is called a **right tailed test**.

Case 2: Testing $\mu_1 < \mu_2$

- Null hypothesis

$$H_0 : \mu_1 \geq \mu_2$$

- Alternative hypothesis

$$H_1 : \mu_1 < \mu_2$$

This is called a **left tailed test**.

The Two-Sample t-Test

Decision rules

For $\mu_1 > \mu_2$

Reject H_0 if:

$$t_0 > t_{\alpha,\, df}$$

For $\mu_1 < \mu_2$

Reject H_0 if:

$$t_0 < -t_{\alpha,\, df}$$

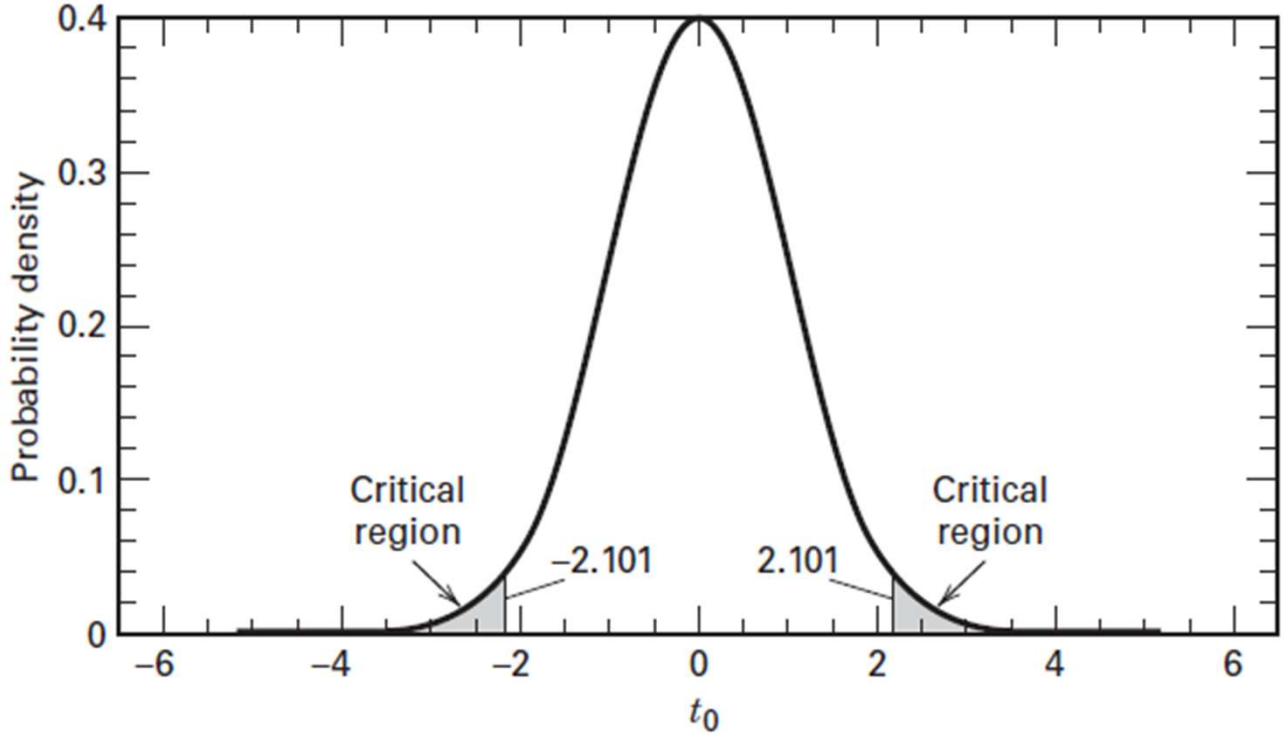
Where:

- α is the significance level
- $df = n_1 + n_2 - 2$

The Two-Sample t-Test

Modified Mortar	Unmodified Mortar
$\bar{y}_1 = 16.76$ kgf/cm^2	$\bar{y}_2 = 17.04$ kgf/cm^2
$S_1^2 = 0.100$	$S_2^2 = 0.061$
$S_1 = 0.316$	$S_2 = 0.248$
$n_1 = 10$	$n_2 = 10$

The Two-Sample t-Test



■ **FIGURE 2.10** The t distribution with 18 degrees of freedom with the critical region $\pm t_{0.025,18} = \pm 2.101$

Use of P-Values in Hypothesis Testing

The *P*-value is the probability that the test statistic will take on a value that is at least as extreme as the observed value of the statistic when the null hypothesis H_0 is true.

Or in other words

The ***P*-value** as the smallest level of significance that would lead to rejection of the null hypothesis H_0

p-value	Interpretation
> 0.10	Data consistent with noise
0.05–0.10	Weak evidence
0.01–0.05	Moderate evidence
< 0.01	Strong evidence

Chi-square distribution

Let

$$Z_1, Z_2, \ldots, Z_k$$

be **independent standard normal random variables**, where

$$Z_i \sim N(0,1)$$

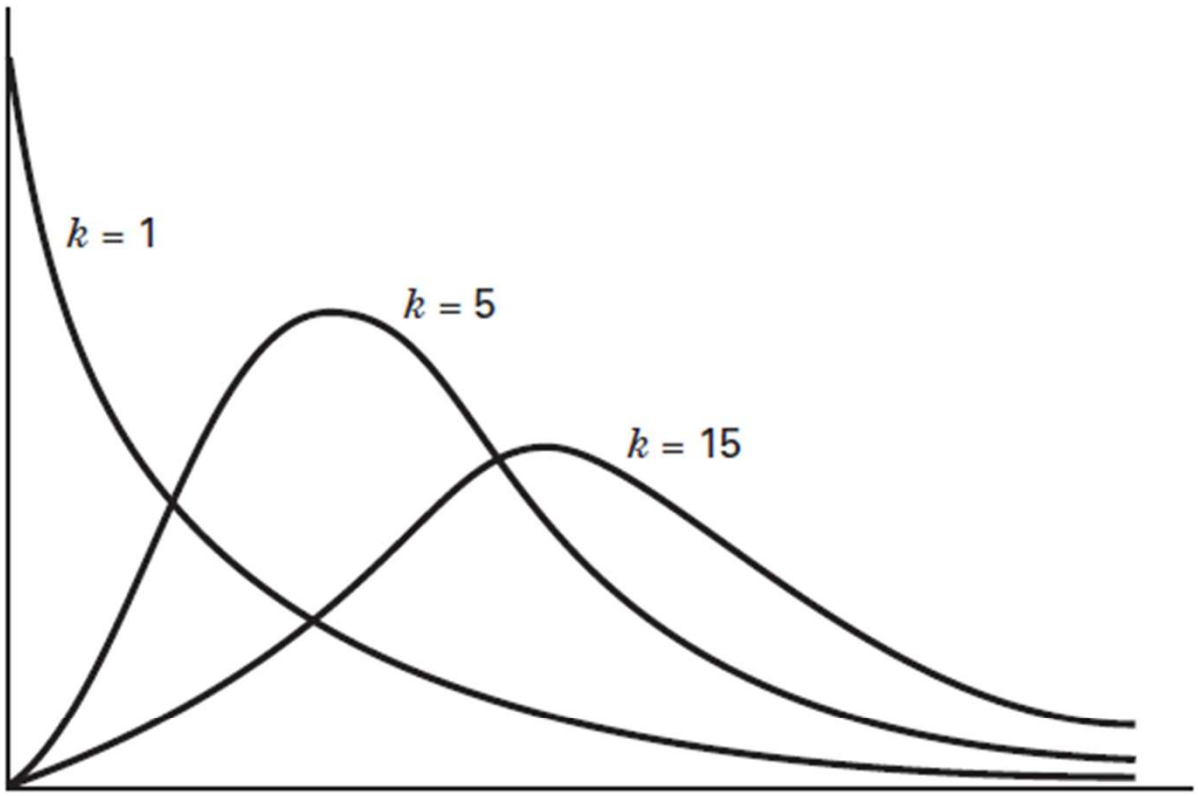
Then the random variable

$$\chi^2 = Z_1^2 + Z_2^2 + \cdots + Z_k^2$$

is said to follow a **chi square distribution with k degrees of freedom**.

$$f(x) = \frac{1}{2^{k/2}\Gamma\left(\frac{k}{2}\right)} x^{(k/2)-1} e^{-x/2} \qquad x > 0$$

Chi-square distribution



■ **FIGURE 2.6** **Several Chi-square distributions**

As the degrees of freedom increase, the chi square distribution becomes approximately normal due to the Central Limit Theorem, since it is formed as a sum of independent squared normal variables.

Chi-square distribution

As an example of a random variable that follows the chi-square distribution, suppose that $y_1, y_2, \ldots, y_n$ is a random sample from an $N(\mu, \sigma^2)$ distribution. Then

$$\frac{SS}{\sigma^2} = \frac{\sum_{i=1}^{n} (y_i - \overline{y})^2}{\sigma^2} \sim \chi^2_{n-1} \tag{2.15}$$

That is, SS/σ^2 is distributed as chi-square with $n - 1$ degrees of freedom.

If the observations in the sample are NID(μ, σ^2), then the distribution of S^2 is $[\sigma^2/(n-1)]\chi^2_{n-1}$. Thus, the sampling distribution of the sample variance is a constant times the chi-square distribution if the population is normally distributed.

If z and χ^2_k are independent standard normal and chi-square random variables, respectively, the random variable

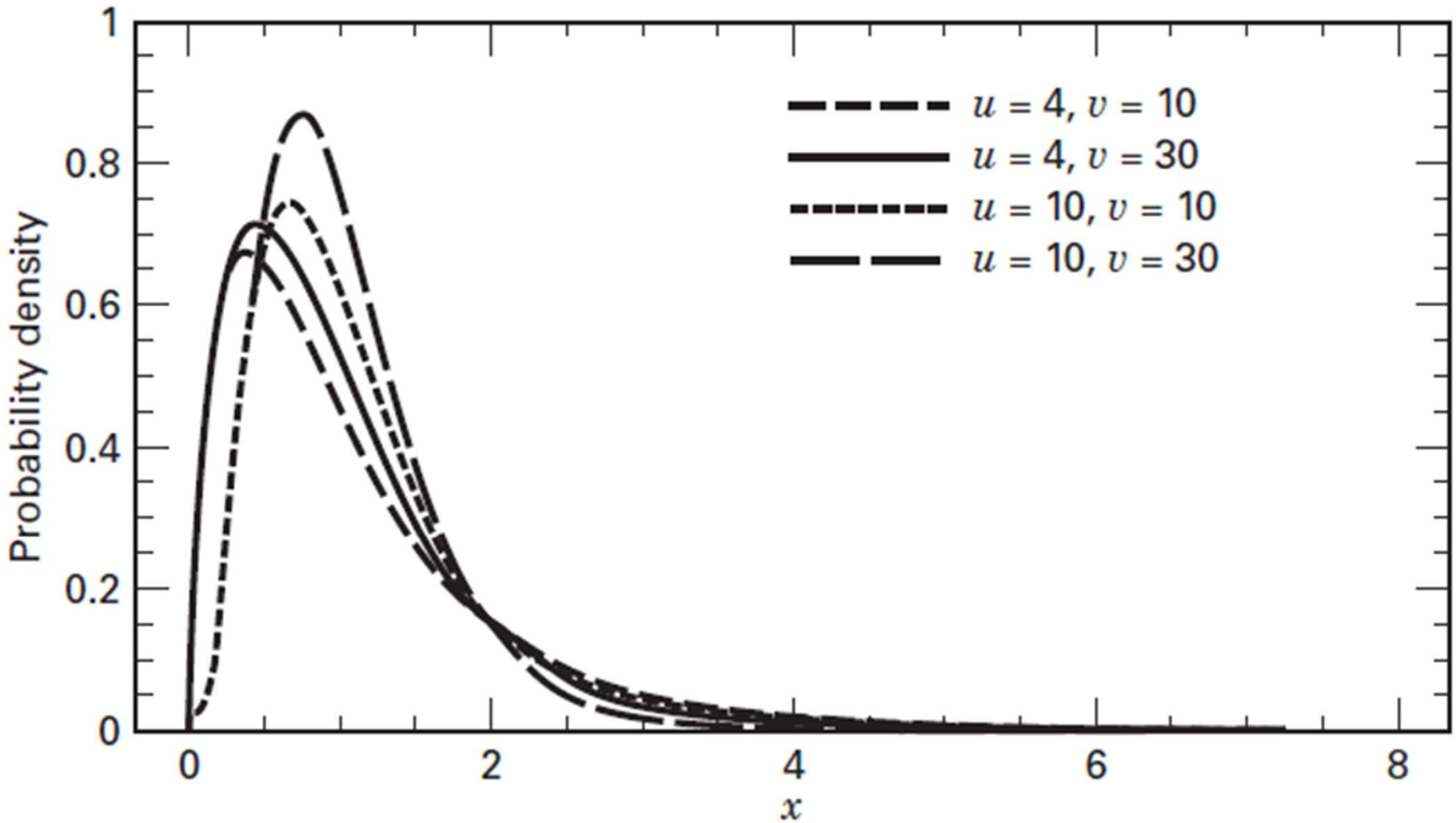
$$t_k = \frac{z}{\sqrt{\chi^2_k/k}} \tag{2.17}$$

F-distribution

The final sampling distribution that we will consider is the ***F* distribution**. If χ_u^2 and χ_v^2 are two independent chi-square random variables with u and v degrees of freedom, respectively, then the ratio

$$F_{u,v} = \frac{\chi_u^2/u}{\chi_v^2/v} \tag{2.20}$$

follows the ***F* distribution with *u numerator* degrees of freedom and *v denominator* degrees of freedom**. If x is an F random variable with u numerator and v denominator degrees of freedom, then the probability distribution of x is

$$h(x) = \frac{\Gamma\left(\frac{u+v}{2}\right)\left(\frac{u}{v}\right)^{u/2} x^{(u/2)-1}}{\Gamma\left(\frac{u}{x}\right)\Gamma\left(\frac{v}{2}\right)\left[\left(\frac{u}{v}\right)x+1\right]^{(u+v)/2}} \qquad 0 < x < \infty \tag{2.21}$$

F-distribution